

CBSE 2027 Maths — Predicted Paper (Set 2 of 10)

Based on the latest CBSE blueprint and a deep analysis of repeating previous year question (PYQ) trends, here is Set 2 of the high-probability predicted Class 12 Maths board paper. This paper is strictly aligned with the official 80-mark, 38-question structure.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

Q1. Let $A = \{1, 2, 3\}$. How many equivalence relations can be defined on set A containing the element $(1, 2)$? * **Chapter:** Relations and Functions * **Probability Tag: Medium**

Q2. Find the principal value of $\sin^{-1}(\sin \frac{2\pi}{3})$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: Very High**

Q3. If A is a square matrix such that $A^2 = A$, then what is the simplified value of $(I + A)^3 - 7A$? * **Chapter:** Matrices * **Probability Tag: High**

Q4. If A is a 3×3 matrix with determinant $|A| = 4$, find the value of $|2A|$. * **Chapter:** Determinants * **Probability Tag: Very High**

Q5. What is the derivative of $\log(\sec x + \tan x)$ with respect to x ? * **Chapter:** Continuity and Differentiability * **Probability Tag: High**

Q6. The radius of a circle is increasing at the rate of 0.7 cm/s . What is the rate of increase of its circumference? * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q7. Evaluate: $\int \frac{1}{\sin^2 x \cos^2 x} dx$. * **Chapter:** Integrals * **Probability Tag: High**

Q8. Find the area of the region bounded by the curve $y = x^3$, the y-axis, and the lines $y = 0$ and $y = 8$. * **Chapter:** Application of Integrals * **Probability Tag: Medium**

Q9. Find the integrating factor of the differential equation: $\frac{dy}{dx} - y = \cos x$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q10. If $\vec{a}$ and $\vec{b}$ are unit vectors and $\vec{a} - \vec{b}$ is also a unit vector, then what is the angle between $\vec{a}$ and $\vec{b}$? * **Chapter:** Vector Algebra * **Probability Tag: Very High**

Q11. Find the scalar (dot) product of the vectors $2\hat{i} - \hat{j} + 3\hat{k}$ and $3\hat{i} + 2\hat{j} - \hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q12. What is the distance of the point $P(a, b, c)$ from the x-axis? * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Medium**

Q13. In a Linear Programming Problem, the optimal value (maximum or minimum) of the objective function is always attained at which part of the feasible region? * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q14. If $P(A) = 0.4$, $P(B) = 0.8$, and $P(B|A) = 0.6$, find $P(A \cap B)$. * **Chapter:** Probability * **Probability Tag: High**

Q15. A matrix $A = [a_{ij}]_{m \times n}$ is said to be a scalar matrix if all its non-diagonal elements are zero and all its diagonal elements are: * **Chapter:** Matrices * **Probability Tag: Medium**

Q16. Evaluate the definite integral: $\int_{-\pi/4}^{\pi/4} x^3 \sin^2 x \, dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q17. Determine the degree of the differential equation: $\left(\frac{dy}{dx}\right)^3 + \frac{d^2y}{dx^2} + \sin\left(\frac{dy}{dx}\right) = 0$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q18. Two standard dice are thrown together. What is the probability of getting a doublet (the same number on both dice)? * **Chapter:** Probability * **Probability Tag: High**

Q19. (Assertion-Reason): Assertion (A): Every scalar matrix is a diagonal matrix. **Reason (R):** A diagonal matrix is a square matrix in which all non-diagonal elements are zero. * **Chapter:** Matrices * **Probability Tag: High**

Q20. (Assertion-Reason): Assertion (A): For any two vectors $\vec{a}$ and $\vec{b}$, $|\vec{a} \cdot \vec{b}| \leq |\vec{a}||\vec{b}|$. **Reason (R):** The angle θ between two vectors always satisfies $-1 \leq \cos \theta \leq 1$. * **Chapter:** Vector Algebra * **Probability Tag: High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

Q21. Simplify the expression: $\tan^{-1} \left[\frac{\sqrt{1+x^2}-1}{x} \right]$, where $x \neq 0$. * **Chapter:** Inverse Trigonometric Functions * **Probability Tag: High**

Q22. Find $\frac{dy}{dx}$ if $y = x^{\sin x}$. * **Chapter:** Continuity and Differentiability * **Probability Tag: Very High**

Q23. Find a vector of magnitude 8 units in the direction of the vector $5\hat{i} - \hat{j} + 2\hat{k}$. * **Chapter:** Vector Algebra * **Probability Tag: High**

Q24. Form the differential equation representing the family of parabolas having their vertex at the origin and axis along the positive y-axis. * **Chapter:** Differential Equations * **Probability Tag: High**

Q25. A family has two children. Find the probability that both are boys, given that at least one of the children is a boy. * **Chapter:** Probability * **Probability Tag: Very High**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Show that the function $f: \mathbb{R} - \{\frac{2}{3}\} \rightarrow \mathbb{R} - \{\frac{2}{3}\}$ defined by $f(x) = \frac{4x+3}{6x-4}$ is a bijective function. * **Chapter:** Relations and Functions * **Probability Tag: High**

Q27. Find the value(s) of k if the area of the triangle with vertices $(2, -6)$, $(5, 4)$, and $(k, 4)$ is 35 sq. units, using determinants. * **Chapter:** Determinants * **Probability Tag: Medium**

Q28. A ladder 5 m long is leaning against a vertical wall. The bottom of the ladder is pulled along the ground, away from the wall, at the rate of **2 cm/s**. How fast is its height on the wall decreasing when the foot of the ladder is 4 m away from the wall? * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q29. Evaluate the integral using partial fractions: $\int \frac{x}{(x-1)(x-2)} dx$. * **Chapter:** Integrals * **Probability Tag: Very High**

Q30. Solve the homogeneous differential equation: $x \frac{dy}{dx} - y = \sqrt{x^2 + y^2}$. * **Chapter:** Differential Equations * **Probability Tag: High**

Q31. Two cards are drawn simultaneously (or successively without replacement) from a well-shuffled pack of 52 cards. Find the probability distribution of the number of aces drawn. * **Chapter:** Probability * **Probability Tag: Very High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. If $A = \begin{pmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{pmatrix}$, find A^{-1} . Using A^{-1} , solve the following system of linear

equations: $2x - 3y + 5z = 11$ $3x + 2y - 4z = -5$ $x + y - 2z = -3$ * **Chapter:** Matrices and Determinants * **Probability Tag: Very High**

Q33. Show that the semi-vertical angle of a right circular cone of maximum volume and of given slant height is $\tan^{-1} \sqrt{2}$. * **Chapter:** Application of Derivatives * **Probability Tag: Very High**

Q34. Find the area of the region enclosed between the two parabolas $y^2 = 4x$ and $x^2 = 4y$. * **Chapter:** Application of Integrals * **Probability Tag: Very High**

Q35. Find the shortest distance between the lines whose vector equations are:
 $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - 3\hat{j} + 2\hat{k})$ $\vec{r} = (4\hat{i} + 5\hat{j} + 6\hat{k}) + \mu(2\hat{i} + 3\hat{j} + \hat{k})$ * **Chapter:** Three-Dimensional Geometry * **Probability Tag: Very High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Linear Programming (Diet Problem) A dietician wishes to mix two types of foods in such a way that the vitamin contents of the mixture contain at least 8 units of vitamin A and 10 units of vitamin C. Food I contains 2 units/kg of vitamin A and 1 unit/kg of vitamin C. Food II contains 1 unit/kg of vitamin A and 2 units/kg of vitamin C. It costs ₹50 per kg to purchase Food I and ₹70 per kg to purchase Food II. Let x be the amount of Food I and y be the amount of Food II. (i) Write the objective function to minimize the cost. (1 Mark) (ii) Formulate the constraints for this Linear Programming Problem. (1 Mark) (iii) Evaluate the corner points of the feasible region and determine the minimum cost of the mixture. (2 Marks) * **Chapter:** Linear Programming * **Probability Tag: Very High**

Q37. Case Study 2: Probability (Bayes' Theorem) In a factory which manufactures bolts and nuts, machines A, B, and C manufacture respectively 25%, 35%, and 40% of the factory's total output of nuts. Of their respective outputs, 5%, 4%, and 2% are defective nuts. A nut is drawn at random from the final product pile and is found to be defective. (i) What is the probability that the drawn nut was manufactured by machine A AND is defective? (1 Mark) (ii) What is the total probability that any randomly drawn nut is defective? (1 Mark) (iii) Given that the drawn nut is defective, find the exact probability that it was manufactured by machine B. (2 Marks) *

Chapter: Probability * **Probability Tag:** Very High

Q38. Case Study 3: Application of Derivatives A telephone company in a town has 500 subscribers on its list and currently collects fixed charges of ₹300 per subscriber per year. The company proposes to increase the annual subscription to generate more profit. Market research shows that for every increase of ₹1 in the annual subscription, exactly one subscriber will discontinue the service. Let x be the increase in the annual subscription fee in rupees. (i) Write the total revenue function $R(x)$ of the company in terms of x . (1 Mark) (ii) Find $\frac{dR}{dx}$ and determine the critical point. (1 Mark) (iii) Use the second derivative test to find the increase in subscription x that yields the maximum revenue, and calculate this maximum revenue. (2 Marks) * **Chapter:** Application of Derivatives * **Probability Tag:** High

CBSE Class 12 (2027) — AI-Predicted via PYQ/Blueprint Analysis. Probability-weighted guidance, not actual exam questions.